ALI NAWAZ

GIS ANALYST | GEOSPATIAL ANALYSIS & REMOTE SENSING

Peshawar, Pakistan | LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

GIS Analyst with 4+ years of experience in zoning, land use planning, spatial analysis, geospatial data management and remote sensing-based research projects. Proficient in ArcMap, ArcGIS Pro, QGIS, Google Earth Engine, and Python-based automation, with expertise in FLUM mapping, parcel data, digitization, georeferencing, and spatial QA/QC. Holds an MS Degree in Geomatics (GIS & Remote Sensing), a Bachelor's degree in Geography, and a PG Diploma in Geomatics.

PROFESSIONAL EXPERIENCE

GIS Analyst — Nook Office Solutions, Islamabad, Pakistan

October 2025 – Present

- Georeferenced zoning maps and imagery using appropriate spatial reference systems and RMS-error control, while performing production QA for topology errors, slivers, gaps, overlaps, and invalid geometries.
- Digitized and quality-assured 1500+ zoning & parcel maps, maintaining 95%+ accuracy across project deliverables.
- Recognized as Employee of Month (August) for consistently high accuracy, productivity and production output.
- Researched and scraped authoritative municipal/county GIS portals, ArcGIS REST Feature Services, and web-based mapping platforms to source current zoning and parcel datasets; identified and documented updated source links for 2000+ cities while maintaining a 92.2% KPI.
- Developed lightweight Python/GIS automation tools to reduce repetitive production tasks, including CRS correction, geometry validation, zone-change detection, sliver and overlap identification, and automated checks for attributes, spelling, missing fields, and geometry gaps.
- Developed a REST spatial-data downloader that accepts GIS service endpoints and automates the extraction of spatial data into local GeoJSON and GeoPackage formats.
- Built a GIS data-extraction and web-scraping utility to streamline the discovery of city-level GIS directories, mapping portals, REST services, and authoritative zoning data sources.
- Worked with PostGIS and QGIS for local spatial database workflows, including importing shapefiles, performing spatial intersections and selections, and managing table inserts, updates, and deletions.
- Utilized raster/TIFF reference datasets to validate zoning symbology and confirm that expected spatial features and geometries were accurately represented.
- Collaborated with GIS and data-science teams to improve data standards, QA procedures, spatial workflows, and production efficiency.
- Conducted QA audits of new team members digitization work, identifying spatial and attribute inconsistencies and providing feedback to improve accuracy and workflow compliance.

GIS Analyst — HP Consultant Planners, Peshawar, KP

October 2023 – October 2025

- Worked on GIS-based master planning and land-use planning projects, helping convert planning data into clear and useful maps.
- Collected and organized spatial information through field surveys and existing datasets to support planning studies.
- Used Geospatial tools to perform spatial analysis, data visualization, and mapping for planning requirements.
- Digitized existing land-use information and prepared LU maps from satellite imagery and other available sources.
- Prepared land suitability maps to support decisions related to future development and land-use planning.
- Developed proposed land-use maps based on planning requirements, existing conditions, and spatial analysis.
- Performed quality checks (QC) on GIS data and maps to identify errors and improve the accuracy of project deliverables.
- Supported the planning team by preparing maps, analyzing spatial data, and providing GIS-based information for planning and decision-making.
- Worked closely with planners and other team members to ensure GIS outputs matched the requirements of ongoing planning projects.

GIS Digitizer — Excise and Taxation Peshawar, KP
March 2023 – September 2023

- Participated in household data collection surveys and helped organize field information for GIS-related work.
- Entered and managed collected data while maintaining consistency and accuracy in records.
- Assisted with image processing and preparation of spatial data for further GIS analysis.
- Digitized geographic features and information from maps and imagery.
- Organized spatial and attribute data to support mapping and departmental requirements.
- Performed basic checks on collected and digitized data to help maintain data quality.

GIS Intern — Board of Revenue Peshawar, KP
January 2023 – February 2023

- Worked on digitization of Masavis maps for Mouza land records.
- Divided Masavi maps into grid sections and digitized cadastral/parcel boundary lines.
- Converted digitized line features into polygons using geospatial tools.
- Added and managed Khasra numbers and related attributes for digitized parcels.
- Performed basic quality checking and correction of digitized cadastral data.

TECHNICAL SKILLS

GIS Software	ArcMap 10.8, ArcGIS Pro, QGIS, Google Earth Engine, ERDAS Imagine, MapInfo.
QA / QC	Shapefile checking, zone field generation, CRS correction, geometry repair, topology validation, zone-change detection, sliver/overlap/gap detection, attribute validation, missing-field checks, GeoJSON export, and automated GIS workflows.
Geospatial Analysis	Zoning and parcel analysis, land-use mapping, cadastral digitization, georeferencing, spatial analysis, spatial data visualization, hazard and flood mapping, suitability & interpolation analysis.
Automation	Python scripting, GIS task automation, repetitive workflow optimization, data processing, and web map data extraction.
Remote Sensing	Landsat, Sentinel-2 multispectral imagery, supervised classification, NDVI, NDBI, NDWI, NDSI indices, FabDEM, SRTM, LULC mapping, time-series analysis, and image stacking.
Programming	Python scripting, GIS task automation, repetitive workflow optimization, data processing, and web map scraping.
Database & Data	PostgreSQL/PostGIS, spatial queries, spatial intersections, table editing, REST Feature Services, GeoJSON, GeoPackage, GPS data, and KoboToolbox.
Soft Skills	Cross-team collaboration, GIS quality auditing, team mentoring, technical documentation, and production workflow improvement.

SELECTED AUTOMATION & GIS PROJECTS

Geospatial QA Automation Suite: Developed Python-based GIS utilities for CRS correction, geometry validation, sliver and overlap detection, field generation, and attribute quality checks to streamline repetitive GIS production workflows.
REST Spatial Data Downloader: Developed a Python-based tool to extract spatial data from ArcGIS REST services and export datasets into GeoJSON and GeoPackage formats for efficient GIS data acquisition.
GIS Portal Discovery & Web Scraping: Built web-scraping utilities to discover municipal GIS portals, REST services, and authoritative zoning and parcel data sources, supporting large-scale GIS production across the USA and Canada.
PostGIS Spatial Database Workflows: Used PostgreSQL/PostGIS and QGIS to manage spatial datasets, perform spatial intersections and selections, and maintain geographic and attribute data.

FEATURED ACADEMIC PROJECTS

Peshawar Valley — Land Surface Temperature & Land Use Analysis: Analyzed land use/land cover (LULC) and land surface temperature (LST) patterns using satellite imagery and GIS to examine spatial and temporal changes across the study area.
Swabi District — LULC Change Detection (2002–2022): Analyzed land-use and land-cover changes over two decades, mapping built-up expansion and spatial changes to support assessment of landscape transformation.
Kurman Valley, Somalia — Flood Susceptibility Analysis: Prepared flood susceptibility and sediment transport maps to identify vulnerable areas and support flood-risk assessment and potential desilting interventions.
Bagh District, AJK Hazard Mapping: Prepared hazard and landslide-susceptibility-related maps using GIS and spatial datasets to assess geographic patterns of potential hazards.
Hackney, UK — School Catchment & Accessibility Mapping: Prepared amenities and services maps as part of a comparative GIS-based study of school catchment areas and active travel.
Caledon, Ontario, Canada — Land Use Map Digitization: Digitized land-use maps to convert existing planning information into structured spatial datasets for GIS analysis and mapping.
Swabi District — Land Use and Built-Up Change Analysis: Mapped changes in built-up areas and land-use patterns using GIS

and remote sensing techniques.

Mouza Taru, Nowshera — Encroachment on State Land: Prepared spatial distribution and overlay maps to examine the geographic distribution of encroachment on state land.

Batura Glacier, Pakistan — Snow Cover Change (2004–2024): Prepared a multi-year snow-cover comparison to visualize temporal changes in glacier-region snow coverage using geospatial data.

"For more details Portfolio link is attached"

EDUCATION

MS Geomatics (GIS & Remote Sensing) — University of Peshawar | 2024–2026

Specialization: Land Surface Temperature and its Impact on Built Environment, Peshawar Valley, KP

BS Geography — University of Peshawar | 2019–2023

Specialization: Urban expansion and growth of Sarai Naurung Lakki Marwat, KP

PG Diploma in Geomatics (GIS/RS) — University of Peshawar | 2023–2024

Specialization: Swabi District Land Use Land Cover Assessment